EEE303 Tutorial 2

1. The following information is given about the simple rotating loop shown in Figure T2-1
   B = 0.8T V_B = 24V
   l = 0.5m R = 0.4 Ω
   r = 0.125m ω = 250 rad/s
   (a) Is this machine operating as a motor or a generator? Explain.
   (b) What is the current i flowing into or out of the machine? What is the power flowing into or out of the machine?
   (c) If the speed of the rotor were changed to 275 rad/s, what would happen to the current flow into or out of the machine?
   (d) If the speed of the rotor were changed to 225 rad/s, what would happen to the current flow into or out of the machine?

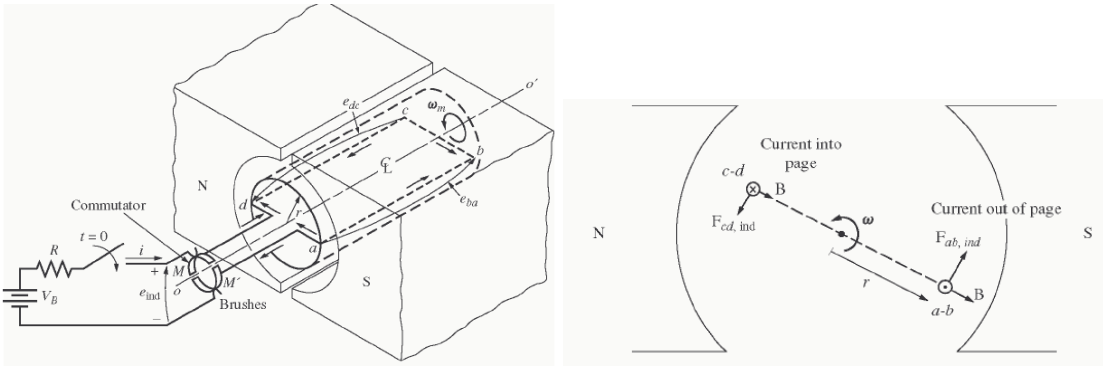


Figure T2-1

2. The armature supply voltage of a DC motor is 230 V. The armature current is 12 A, the armature resistance is 0.8 Ω, and the speed is 100 rad/s (Figure T2-2). Calculate
   (a) the induced emf
   (b) the electromagnetic torque
   (c) the electrical power input to the armature
   (d) the mechanical power developed by the armature
   (e) the armature copper-losses?

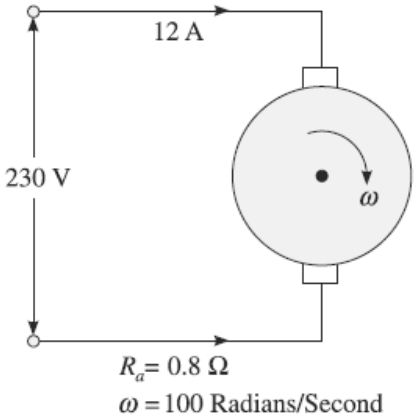


Figure T2-2

3. A 240 V DC shunt motor runs on no-load at 800 rpm with no extra resistance in the field or armature circuit, the armature current being 2 A. Calculate the resistance required in series with the shunt winding so that the motor may run at 950 rpm when taking a line current of 30

A. Shunt winding resistance is 160 Ω, armature resistance is 0.4 Ω, Assume that flux is proportional to field current.

4. A 230 V DC shunt motor takes an armature current of 20 A on a particular load. The armature circuit resistance is 0.5 Ω. Find the resistance required in series with the armature to reduce the speed by 50%, if

   (a) the load torque is constant

   (b) the load torque is proportional to the square of the speed.

5. The resistance of the armature circuit of a 250 V DC shunt motor is 0.3 Ω and its full-load speed is 1000 rpm. Calculate

   (a) the resistance required in series with the armature to reduce the speed with the full-load torque to 800 rpm, the full-load armature current being 50 A.

   (b) If the load torque is then halved, at what speed will the motor run? The armature reaction effect is to be neglected.

6. A 4 pole, 500 V, shunt motor has a total of 720 armature conductors which are wave connected. The full-load armature current is 60 A, and the flux per pole is 0.03 Wb. The armature resistance is 0.2 Ω. The voltage drop across a brush is 1 volt. Calculate the full-load speed of the motor.

7. A DC series motor (Figure T2-3) runs at 1000 rpm when taking 20 A at 200 V. The resistance of the armature circuit is 0.5 Ω and that of the field winding is 0.2 Ω. Find the speed for a total current of 20 A, 200 V, when a 0.2 W resistor is joined in parallel with the field winding. The flux for a field current of 10 A is 70% of that for 20 A.

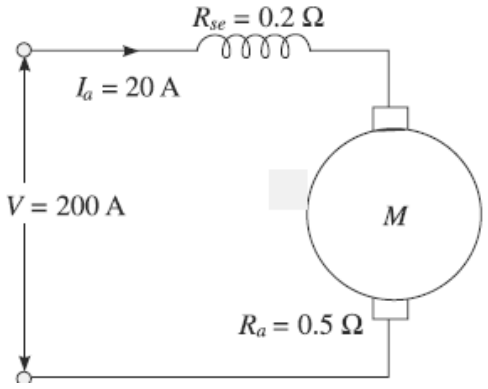


Figure T2-3

8. If the resistor R_{adj} in the shunt DC motor (Figure T2-4) is adjusted to 175 Ω, calculate

   (a) the field current I_F under no-load conditions, where $V_T = E_A$.

   (b) the speed under no-load conditions, assuming at full load, the back emf (E_A or E_b) is 271V and the speed is 1200 rpm based on I_F above.

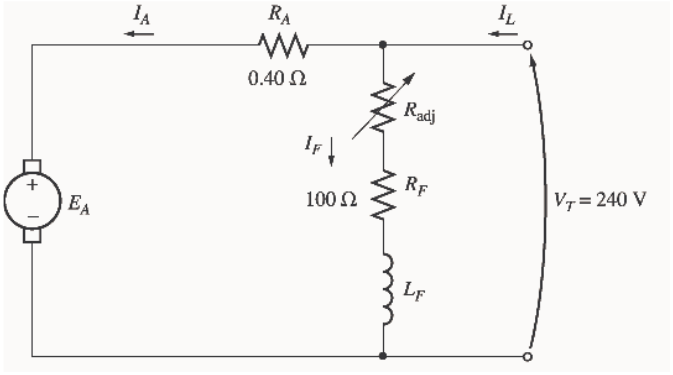


Figure T2-4

9. If the resistor R_{adj} in the separately excited DC motor (Figure T2-5) is adjusted to 175 Ω, calculate

   (a) the field current I_F under no-load conditions, where $V_A = E_A$.

   (b) the no-load speed of this separately excited motor when R_{adj} = 175 Ω and (i) V_A = 120 V, (ii) V_A = 180 V, (iii) V_A = 240 V? Assuming at full load, the back emf (E_A or E_b) is 271V and the speed is 1200 rpm based on I_F above.

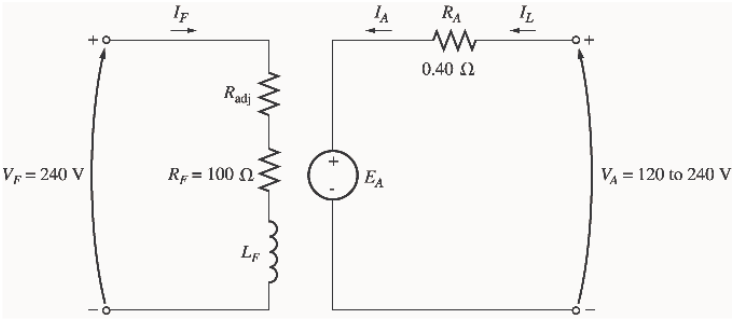


Figure T2-5

10. A 7.5-hp 120-V series dc motor has an armature resistance of 0.2 Ω and a series field resistance of 0.16 Ω. At full load, the current input is 58 A, and the rated speed is 1050 rpm. Its magnetization curve is shown in Figure T2-6. The core losses are 200 W, and the mechanical losses are 240 W at full load. Assume that the mechanical losses vary as the cube of the speed of the motor and that the core losses are constant.

    (a) What is the efficiency of the motor at full load?

    (b) What are the speed and efficiency of the motor if it is operating at an armature current of 35 A?

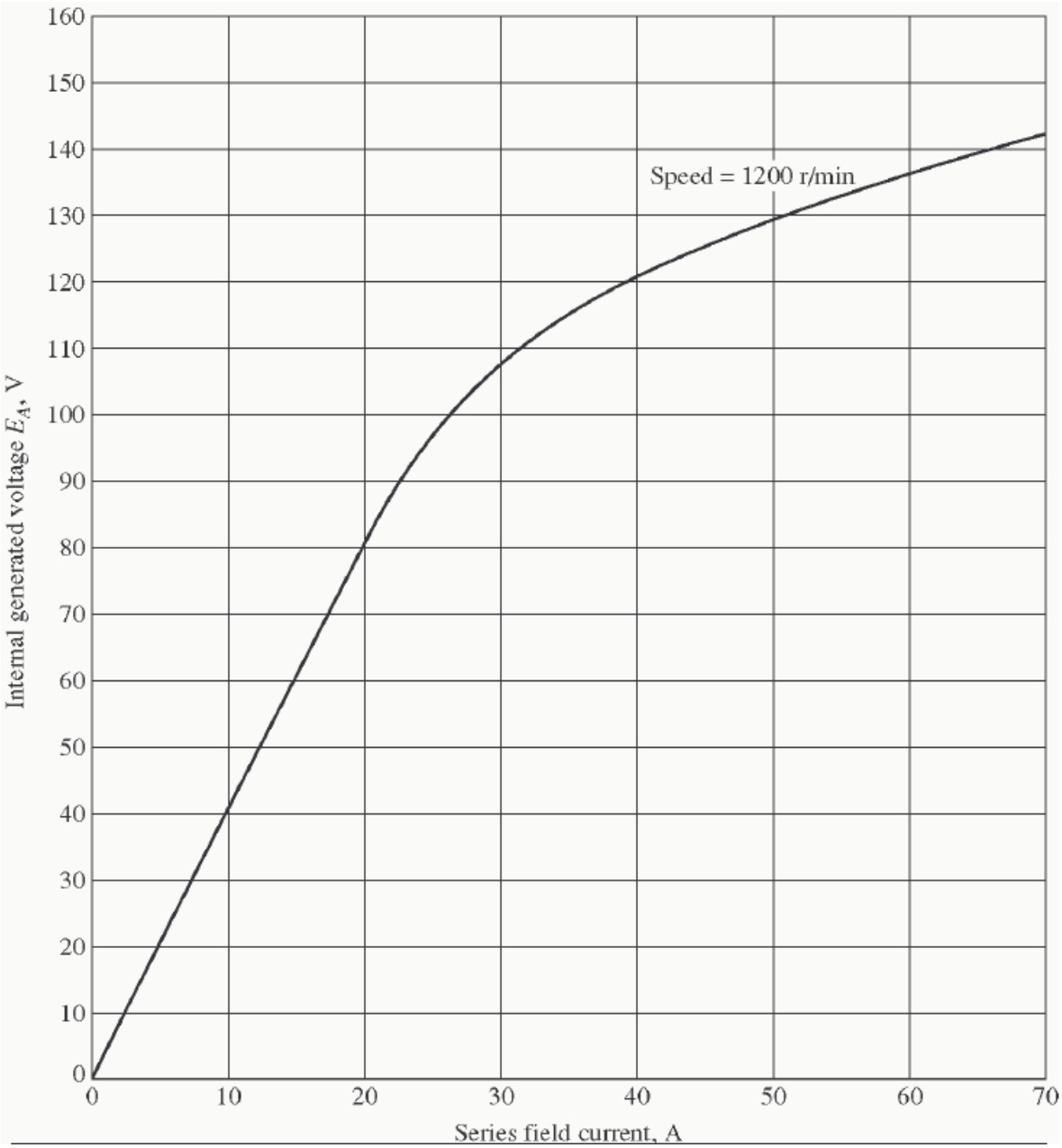

Internal generated voltage E_A, V
Series field current, A
Speed = 1200 r/min
0
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150
160

Figure T2-6